### Integrating **wildfire spread** and evacuation times to design safe triggers: Application to two rural communities using PERIL **model**

H Mitchell, S Gwynne, E Ronchi, N Kalogeropoulos… - Safety science, 2023 - Elsevier

… the **wildfire** indicate it is time to evacuate. Triggers are often arbitrarily defined via identifiable landmarks, do not include a safety factor, and do not account for fire **spread** … of **wildfire** and …

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R Sun, SK Krueger, MA Jenkins… - … of **Wildland Fire**, 2009 - CSIRO Publishing

… The major source of uncertainty in **wildfire behavior prediction** is the transient **behavior** of **wildfire** due … The coupled **wildfire**–atmosphere **model** used in the present study is limited by the …

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### **Review** of **wildfire** modeling considering effects on land surfaces

D Or, E Furtak-Cole, M Berli, R Shillito… - Earth-Science …, 2023 - Elsevier

… **review** are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances in modeling of **wildfire dynamics** and rates of **spread** … large eddy **simulation** models. We …

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### **Review** of **Wildfire** Detection, Fighting, and Technologies: Future Prospects and Insights

NM Negash, L Sun, C Fan, D Shi, F Wang - AIAA AVIATION FORUM …, 2025 - arc.aiaa.org

… management tasks depicted in the framework include fire detection, **wildfire** segmentation, **spread prediction**, and risk assessment, aligning with the capabilities commonly emphasized …

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H Hesseln - Current Forestry Reports, 2018 - Springer

… **behavior** and factors that will lead to enhanced **wildfire** … methods to **model** the likelihood of **wildfire** including those … spatio-temporal dimensions and **model** outputs. Their findings …

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### Coupling wind **dynamics** into a DDDAS **forest fire propagation prediction** system

C Brun, T Artées, T Margalef, A Cortées - Procedia Computer Science, 2012 - Elsevier

… **prediction** quality improvements when coupling **forest fire spread model** with weather **forecast** … Rothermel fire **propagation model**[14], has been used as **forest fire spread** simulator and, …

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### The distributed strategy for asynchronous observations in data-driven **wildland fire spread prediction**

M Zha, Z Wang, J Ji, J Zhu - International Journal of **Wildland Fire**, 2024 - CSIRO Publishing

… as the basic fire **propagation model**, which is widely used for **wildfire prediction**. Real terrain, fuel … the **prediction**. Fuel adjustment factors and wind parameters were randomly generated. …

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X Hu, Y Sun, L Ntaimo - **Simulation**, 2012 - journals.sagepub.com

… **simulation model** and uses spatial fire **behavior** input data on topography and the Canadian **forest fire behavior prediction** (… to modeling the **spread** of **wildland fire**: A **review**. Prog Phys …

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Q Huang, A Razi, F Afghah… - 2020 IEEE 21st …, 2020 - ieeexplore.ieee.org

… We evaluate the fire **propagation model** by comparing the results with the … modified and complements the Rothermel’s **model**, we first **review** this **model**, and then mention how we …

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KRA Pillai, YEAD Sultan - International Journal of Innovative …, 2023 - researchgate.net

… of **wildfire** modeling is to understand and **forecast** fire **behavior** using numerical **simulation** … Past models of **wildfire spread** have included basic ellipses, egg- and fan-shaped models, …

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